DICT312 ASSIGNMENT 1

GROUP PROJECT ASSIGNMENT

PROPOSAL FOR AN INTEGRATED CAR RENTAL MANAGEMENT SYSTEM

Pace Car Rental – Mbombela/Nelspruit

Project Information	Details
Service Provider	Red-Code Company LTD
Client / Customer	Pace Car Rental – Mbombela/Nelspruit
Proposed System	Pace Integrated Car Rental Management System (PICRMS)
Course	DICT312
Group Name	Future Developers
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Table of Contents

1. Introduction
2. Client Organisation
3. Service Provider – Red-Code Company LTD
4. Stakeholders and Consultation
5. Task Given by the Client
6. Group Approach
7. Proposed Solution – PICRMS
8. UML and System Analysis
9. Database and Data Management
10. Database Implementation and Sample Data
11. Java and Jakarta EE Technologies
12. Transaction Points and Business Logic
13. Automation Opportunities
14. Dashboard, Reporting and Social-Media Connectivity
15. Security Management
16. Expected Benefits and Value to the Client
17. Conclusion

1. Introduction

Modern car-rental operations depend on accurate information about vehicles, customers, reservations, rentals, payments and maintenance. A rental transaction can involve several employees and several pieces of information, so a centralised information system can reduce duplication and make the status of each transaction easier to understand.

This proposal uses Pace Car Rental in Mbombela/Nelspruit as the client context. Pace's official location information identifies its Mbombela/Nelspruit branch at 14 Amanzi Street, Riverside Park Ext 24, and lists a range of rental services. The official site also provides online booking functionality.

The project does not claim that Pace currently has any specific internal weakness unless such a weakness is confirmed through consultation. Instead, the proposal identifies business requirements and opportunities that should be validated with the client. This distinction is important because the final implemented solution should be based on confirmed requirements rather than assumptions.

1.1 Purpose of the Proposal

The purpose is to recommend a practical enterprise system architecture and implementation approach that satisfies the assignment requirements while demonstrating how a real car-rental business could manage its activities through one integrated platform.

1.2 Scope

- Customer registration and profile management
- Vehicle and fleet management
- Booking and rental management
- Payment, invoice and receipt records
- Vehicle maintenance records
- Dashboard and reporting
- Notifications and reminders
- Role-based security and audit logging
- Five-year record retention and archiving
- Social-media connectivity points

1.3 Out of Scope for the Initial Prototype

- Physical vehicle telematics and GPS hardware integration
- Automatic vehicle door unlocking
- Direct banking infrastructure
- Production payment-gateway credentials
- Access to Pace's private social-media accounts

2. Client Organisation

The selected client is Pace Car Rental – Mbombela/Nelspruit. The official Pace website describes the Mbombela/Nelspruit operation as a car-rental service serving local and visiting customers in the Lowveld. The published location page provides the branch address, contact information and operating hours, and presents multiple vehicle categories and rental options.

2.1 Client Services Relevant to the Project

Service area	System relevance
Vehicle rental	Requires vehicle availability, pricing and status tracking.
Customer bookings	Requires customer and reservation records.
Vehicle collection and return	Requires rental lifecycle and return processing.
Fleet maintenance	Requires service history and maintenance status.
Payments and rental charges	Requires transaction and balance records.
Customer communication	Requires confirmations, reminders and contact channels.

2.2 Business Units / Departments Considered

Unit	Responsibilities
Management	Business oversight, reports, revenue and utilisation.
Rental / Front Desk	Customers, bookings, collections, returns and payments.
Fleet Operations	Vehicle availability, inspections and maintenance.
Administration	User accounts, records, reporting and audit activities.
Customers	Registration, vehicle search, booking and payment information.

3. Service Provider – Red-Code Company LTD

Red-Code Company LTD is presented in this academic project as the local software-services provider responsible for analysing requirements, designing the proposed architecture, developing the prototype and recommending an implementation plan. The company is treated as the student consulting/development organisation for purposes of the assignment.

3.1 Services Offered by Red-Code

- Business and systems analysis
- Web application development
- Database design and implementation
- Java/Jakarta EE development
- System integration and REST API development
- Cybersecurity and access-control design
- Testing and quality assurance

- Deployment and technical support
- User training and documentation

3.2 Proposed Delivery Responsibilities

Phase	Red-Code responsibility
Discovery	Interview stakeholders and document requirements.
Analysis	Map current processes and identify system requirements.
Design	Create architecture, UML, database and interface designs.
Development	Implement the Java/Jakarta EE web application.
Testing	Perform functional, integration, security and usability tests.
Deployment	Configure application server, database and production environment.
Support	Provide maintenance, backups, monitoring and user support.

4. Stakeholders and Consultation

Stakeholder consultation is necessary because the assignment asks the software team to advise the client on how the desired platform can be achieved. The following stakeholder groups should be consulted before implementation.

Stakeholder	Information needed	Consultation method
Branch Manager	KPIs, reports, business rules and approval processes	Interview / workshop
Rental Staff	Booking, customer, collection and return procedures	Observation / interview
Fleet Officer	Vehicle status, inspection and maintenance processes	Interview / process mapping
Finance/Admin	Payments, invoices, reconciliation and retention	Interview / document review
Customers	Booking experience and communication needs	Questionnaire / usability session
IT / Technical support	Hosting, networking, backups and integration	Technical workshop
Project Team	Requirements, design, coding and testing	Agile meetings

4.1 Consultation Questions

1. How are vehicle availability and reservations currently recorded?
2. What information is required before a customer can rent a vehicle?
3. How are vehicle collections and returns processed?
4. How are payments, deposits and outstanding balances recorded?
5. How are vehicles scheduled for maintenance?
6. What reports does management need daily, weekly and monthly?
7. Who should be allowed to view or change customer information?
8. How long must operational and financial records be retained?
9. What notifications should be automatic?
10. Which social-media channels should be linked or integrated?

4.2 Problem Analysis – Fishbone Diagram

The following fishbone diagram provides a proposed analysis of information-management factors that the PICRMS requirements should address. These factors are proposed for validation during stakeholder consultation; they are not claims about confirmed weaknesses at Pace Car Rental.

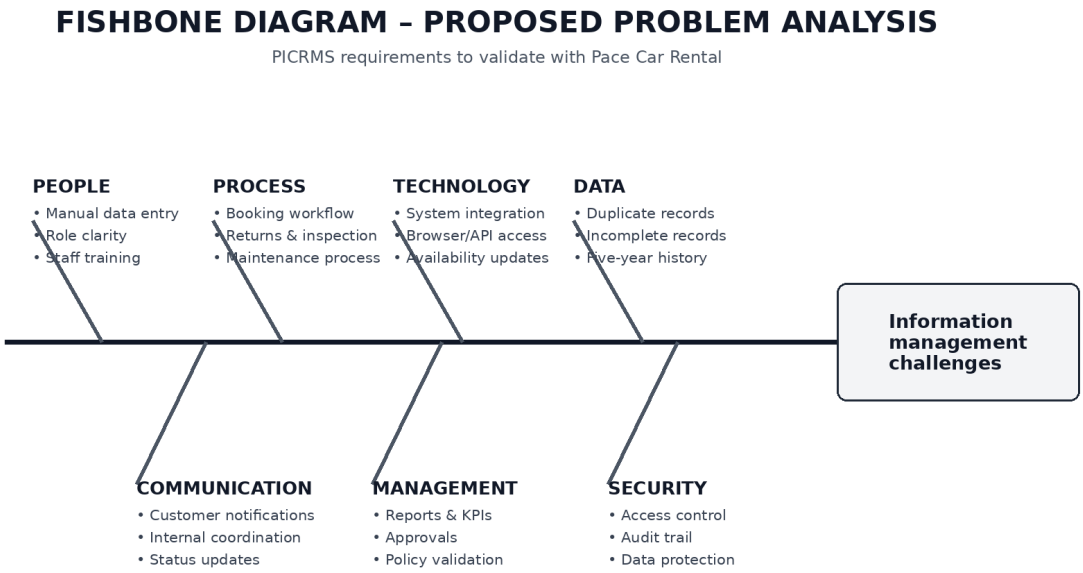


Figure 2: Proposed fishbone problem analysis for PICRMS

5. Task Given by the Client

The proposed task is to design and develop an integrated platform that brings major rental-business activities into one system. The platform should allow authorised users to manage activities from a dashboard/console, retain business records for at least five years, be accessible through a browser, connect to appropriate social-media channels, and use Java/Jakarta technologies.

5.1 Three Core Aspects

Aspect	Core functions	Expected benefit
1. Vehicle & Fleet Management	Vehicles, status, mileage, rates, availability and maintenance	Accurate fleet visibility and fewer availability errors.
2. Customer & Booking Management	Registration, profiles, availability search, bookings and rental history	Faster reservations and better customer service.
3. Payment & Rental Management	Charges, deposits, payments, invoices, receipts and balances	Accurate financial tracking and easier reconciliation.

5.2 Integrated Transaction Example

A single customer transaction may start when a customer searches for an available vehicle. The booking module checks the vehicle calendar, creates a reservation and calculates the rental cost. When the rental is collected, the vehicle status changes to RENTED. A payment is linked to the booking. On return, staff record the vehicle condition and the system updates the vehicle status. If maintenance is required, the maintenance module records the work before the vehicle becomes available again.

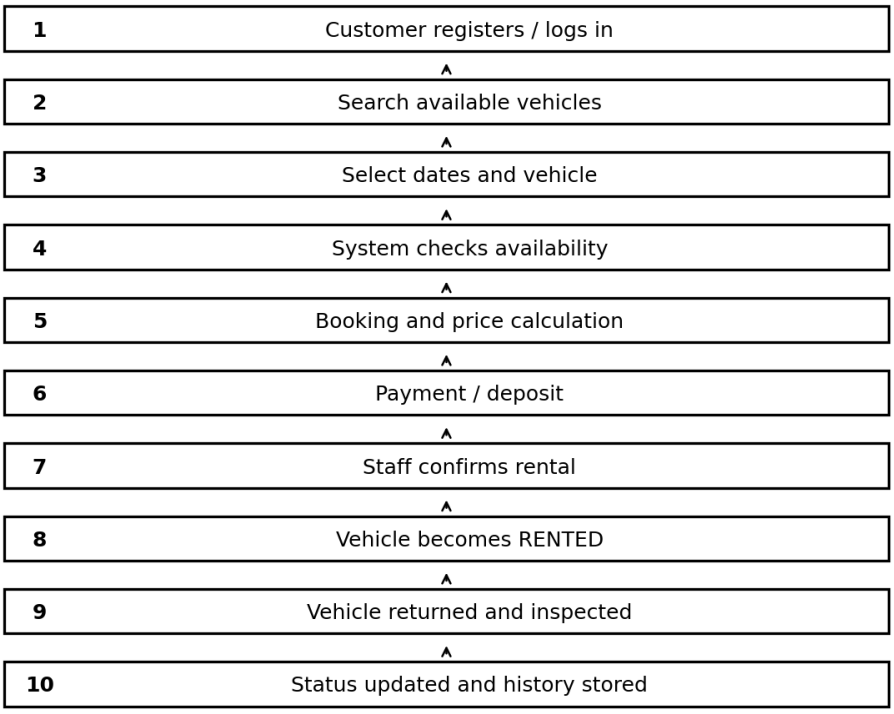


Figure 1: End-to-end rental transaction workflow

6. Group Approach 9

Our team will use an iterative Agile-inspired approach. Our team will first confirm requirements, then produce a minimum viable prototype and improve it through testing and stakeholder feedback. This approach reduces the risk of designing a system around assumptions that have not been validated.

6.1 Project Phases

Phase	Activities	Deliverables
1. Discovery	Stakeholder interviews and requirement gathering	Requirement notes
2. Analysis	Process mapping and data analysis	Functional/non-functional requirements
3. Design	Architecture, UML, ERD and interface design	Design package
4. Development	Java/Jakarta EE and database implementation	Working prototype
5. Testing	Functional, integration, security and usability tests	Test evidence
6. Deployment	Server/database configuration	Deployed system
7. Evaluation	User feedback and refinement	Evaluation report

6.2 Agile Roles

Role	Responsibility
Project coordinator	Coordinates milestones and communication.
Business analyst	Documents requirements and processes.
Backend developer	Implements Java/Jakarta EE services.
Database developer	Designs MySQL schema and queries.
Frontend developer	Builds responsive browser interfaces.
QA / tester	Creates and executes test cases.
Documentation lead	Maintains proposal, diagrams and user documentation.

7. Proposed Solution – PICRMS

The proposed solution is the Pace Integrated Car Rental Management System (PICRMS). It is a browser-based, role-aware business application built around a central relational database. The platform will integrate operational modules so that information entered at one transaction point can be reused by other authorised modules.

7.1 Main Modules

- Authentication and user management
- Customer management
- Vehicle and fleet management
- Booking management
- Rental collection and return
- Payment and invoice management
- Maintenance management
- Notifications
- Dashboard and reporting
- Audit logging
- Social-media/contact integration

7.2 System Architecture

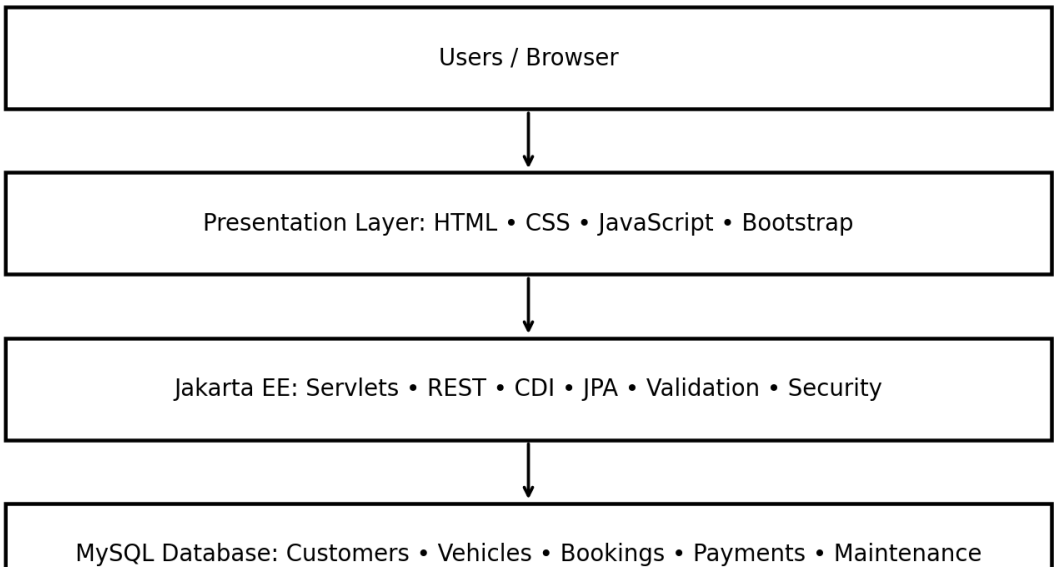


Figure 5: Proposed three-tier PICRMS architecture

The presentation layer provides browser interfaces. The application layer contains Jakarta EE services and business rules. The data layer stores structured information in MySQL. This separation improves maintainability because user-interface changes can be made without rewriting the entire data layer.

7.3 Integrated Client / API Architecture

The following architecture complements the three-tier view by showing how browser and mobile clients can communicate with the PICRMS back-end through controlled API connectors. The mobile interface is presented as a proposed extension; the core prototype remains a responsive browser application.

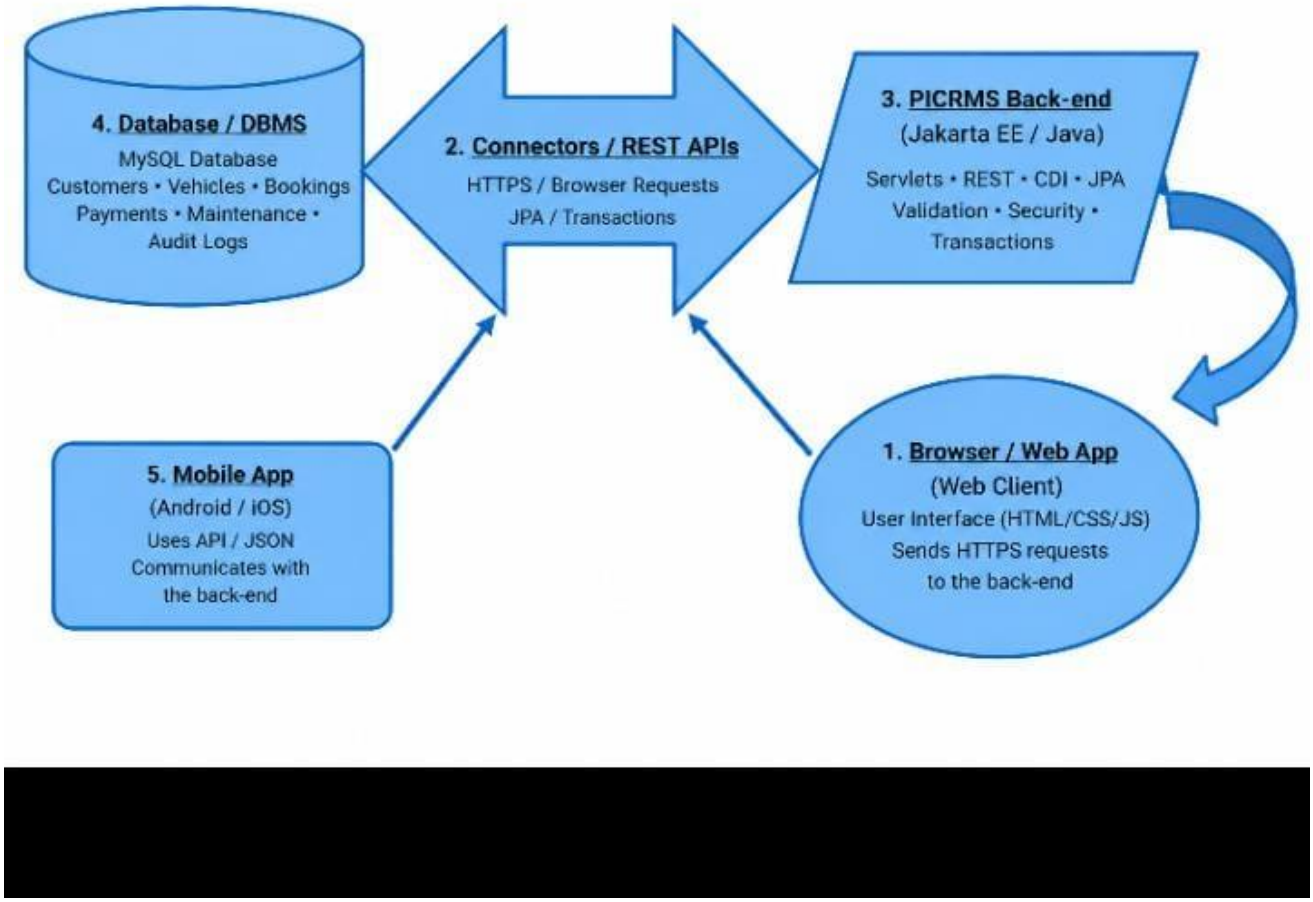


Figure 3: Proposed integrated client, API and back-end architecture

7.4 Proposed Home Page

The proposed home page provides the public entry point to PICRMS. It includes navigation to vehicles, information and contact pages, together with clear actions for searching vehicles and starting a booking. The design is a demonstration mock-up and should be refined after client feedback.

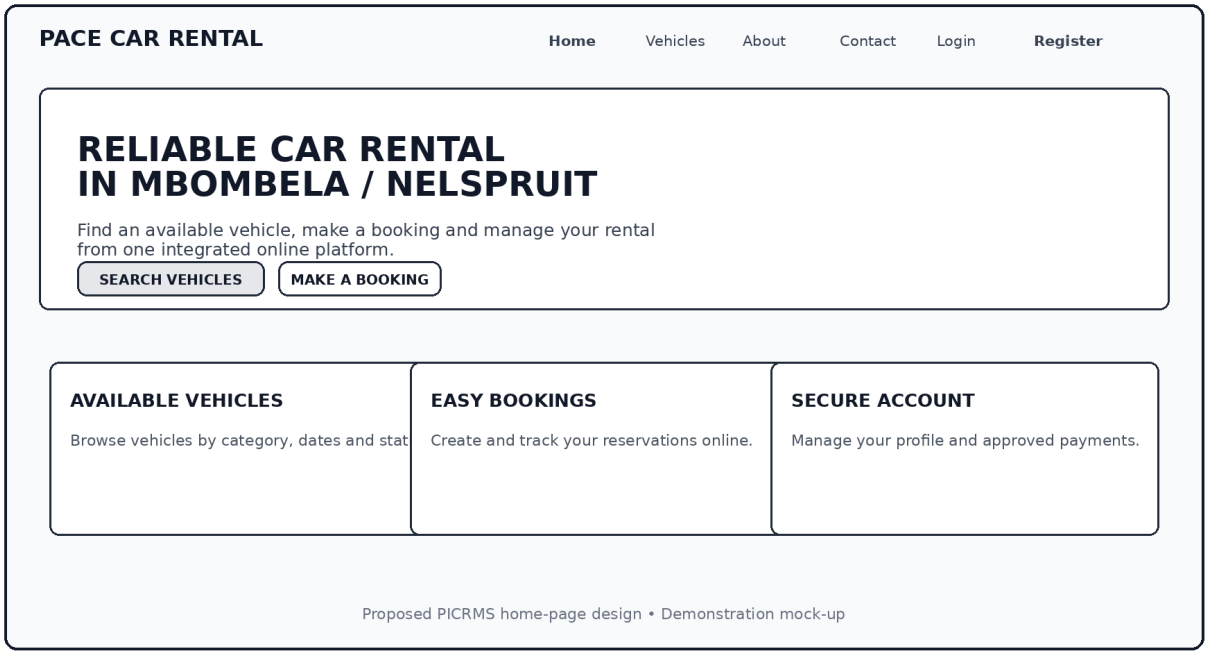


Figure 4: Proposed PICRMS home-page design

8. UML and System Analysis

8.1 Use-Case Diagram

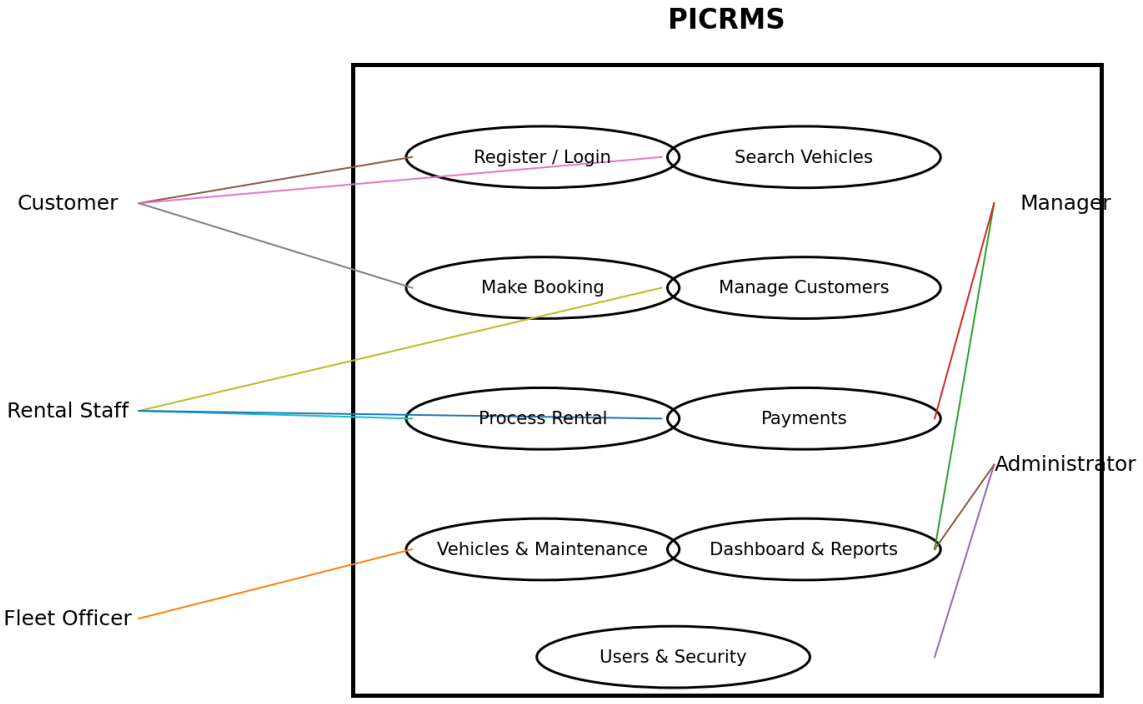


Figure 6: High-level PICRMS use-case diagram

The use-case model identifies the main actors and the actions they perform. Customers can register, search vehicles and create bookings. Rental staff manage customers, bookings, rentals and payments. Fleet officers manage vehicles and maintenance. Managers use dashboards and reports. Administrators manage users, roles and system configuration.

8.2 Key Use Cases

Use case	Actor	Main outcome
Register / Login	Customer, Staff	Authenticated user session.
Search Vehicles	Customer, Staff	List of vehicles matching availability criteria.
Make Booking	Customer, Staff	Confirmed or pending reservation.
Process Rental	Staff	Vehicle assigned to active rental.
Return Vehicle	Staff / Fleet Officer	Rental closed and vehicle status updated.
Record Payment	Staff	Payment linked to booking and balance recalculated.
Manage Maintenance	Fleet Officer	Service event recorded.
View Dashboard	Manager / Admin	Operational and management KPIs.
Manage Users	Admin	Accounts and permissions controlled.

8.3 Booking Business Rule

```
IF a selected vehicle has an existing active booking
AND the requested rental dates overlap
THEN reject the new booking
ELSE create the booking and calculate the rental amount.
```

9. Database and Data Management

The database is the central repository for operational records. A relational design is appropriate because the business contains clear entities and relationships: a customer can have many bookings, a vehicle can be involved in many bookings over time, a booking can have payments, and a vehicle can have multiple maintenance events.

9.1 Entity Relationship Diagram

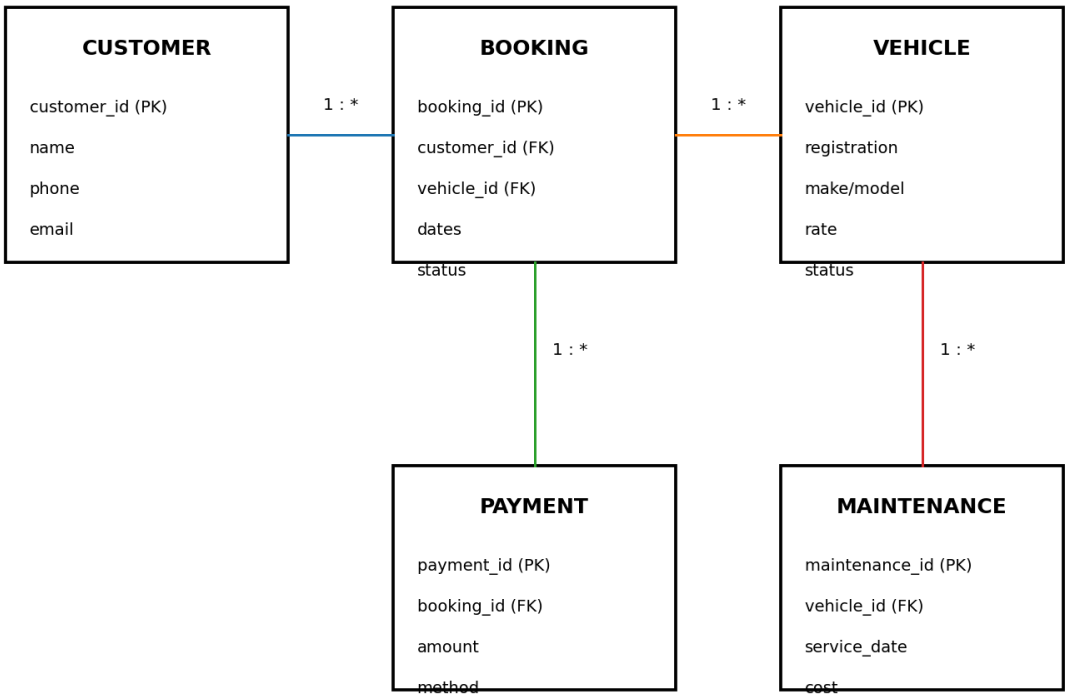


Figure 7: Simplified entity relationship diagram

9.2 Main Tables

Table	Purpose	Key fields
users	System accounts and roles	user_id, username, role, password_hash
customers	Customer profiles	customer_id, name, ID/passport, phone, email
vehicles	Fleet information	vehicle_id, registration, make, model, rate, status
vehicle_types	Vehicle categories	type_id, name, description
bookings	Reservations	booking_id, customer_id, vehicle_id, dates, status
rentals	Active/completed rental events	rental_id, booking_id, collection, return
payments	Financial transactions	payment_id, booking_id, amount, method

invoices	Billing documents	invoice_id, booking_id, total, issue_date
maintenance	Vehicle service history	maintenance_id, vehicle_id, service_date, cost
notifications	Messages and reminders	notification_id, recipient, type, status
audit_logs	Security and accountability	log_id, user_id, action, timestamp

9.3 Customer Data Stored

- Name and contact information
- Identification/passport reference where required by business policy
- Driver's licence information where required
- Address
- Booking and rental history
- Payment references and balances
- Communication preferences

10. Database Implementation and Sample Data

MySQL is proposed for the relational database. MySQL 8.4 documentation describes the server as a multi-user SQL database server and provides documentation for security, backup/recovery, transactions, indexes and other administration topics.

10.1 Sample Database Definition

```
CREATE DATABASE pace_car_rental;
USE pace_car_rental;

CREATE TABLE customers (
  customer_id BIGINT AUTO_INCREMENT PRIMARY KEY,
  first_name VARCHAR(100) NOT NULL,
  last_name VARCHAR(100) NOT NULL,
  id_number VARCHAR(50) UNIQUE NOT NULL,
  phone VARCHAR(30),
  email VARCHAR(150),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE vehicles (
  vehicle_id BIGINT AUTO_INCREMENT PRIMARY KEY,
  registration_number VARCHAR(30) UNIQUE NOT NULL,
  make VARCHAR(100) NOT NULL,
  model VARCHAR(100) NOT NULL,
  daily_rate DECIMAL(10,2) NOT NULL,
  status VARCHAR(30) DEFAULT 'AVAILABLE'
);

CREATE TABLE bookings (
  booking_id BIGINT AUTO_INCREMENT PRIMARY KEY,
  customer_id BIGINT NOT NULL,
  vehicle_id BIGINT NOT NULL,
  pickup_date DATE NOT NULL,
  return_date DATE NOT NULL,
  total_amount DECIMAL(10,2),
  booking_status VARCHAR(30) DEFAULT 'PENDING'
);
```

10.2 Sample Data

Customer ID	Customer	Vehicle	Booking dates	Status
C001	Thabo Mokoena	Toyota Starlet	10–13 Sep 2026	CONFIRMED
C002	Naledi Maseko	Suzuki Swift	12–15 Sep 2026	PENDING
C003	Sibusiso Dlamini	Toyota Corolla Quest	16–20 Sep 2026	CONFIRMED
C004	Lerato Nkosi	Toyota Vitz	21–23 Sep 2026	COMPLETED
C005	Mandla Sithole	Suzuki Ciaz	24–28 Sep 2026	CANCELLED

Vehicle	Sample daily rate	Sample status	Maintenance
Toyota Starlet	R705	AVAILABLE	Next service due
Suzuki Swift	R540	RENTED	No immediate action
Toyota Vitz	R640	AVAILABLE	Serviced
Suzuki Ciaz	R675	MAINTENANCE	Brake inspection
Toyota Corolla Quest	R755	AVAILABLE	Serviced

All names, booking details and sample rates in the tables above are fictional demonstration data and should not be presented as actual Pace customer records or current quotations.

11. Java and Jakarta EE Technologies

Jakarta EE is recommended because it provides standard enterprise APIs for building secure, maintainable Java applications. Jakarta EE 11 is a current platform release and requires Java SE 17 or higher. The selected APIs are matched to specific transaction points in PICRMS.

Technology	Use in PICRMS	Value
Jakarta Servlets	Handle browser requests and responses	Reliable server-side web processing.
Jakarta RESTful Web Services	Expose REST endpoints for vehicles, bookings and payments	Supports modular API integration.
Jakarta Persistence (JPA)	Map Java entities to MySQL tables	Reduces repetitive database access code.
CDI	Inject services and manage application components	Loose coupling and maintainability.
Bean Validation	Validate customer, vehicle and booking fields	Improves data quality.
Jakarta Security	Authentication, authorisation and role-based access	Protects business and customer data.
Jakarta Transactions	Manage multi-step database operations	Helps preserve data consistency.
Jakarta JSON Processing / Binding	Handle JSON data exchanged through APIs	Supports browser/API communication.

11.1 Example Entity

```
@Entity
@Table(name = "vehicles")
```

```
public class Vehicle {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long vehicleId;

    @NotBlank
    private String registrationNumber;

    @NotBlank
    private String make;

    @NotBlank
    private String model;

    private double dailyRate;
    private String status;
}
```

11.2 Example REST Resource

```
@Path("/vehicles")
@Produces(MediaType.APPLICATION_JSON)
public class VehicleResource {

    @GET
    public List<Vehicle> getVehicles() {
        return vehicleService.getAllVehicles();
    }
}
```

12. Transaction Points and Business Logic

A transactional point is a stage where the system receives, validates, stores or changes business information. PICRMS will apply business rules at these points so that the platform does more than store data.

Transaction point	Rule / automation	Result
Vehicle search	Filter by date, category and status	Only suitable vehicles shown.
Booking creation	Check date overlap	Double bookings prevented.
Price calculation	Days × daily rate + approved charges	Rental total calculated.
Payment	Validate amount and booking status	Balance updated.
Collection	Confirm booking and customer requirements	Vehicle becomes RENTED.
Return	Record return date and inspection	Vehicle becomes RETURNED / MAINTENANCE.
Maintenance	Flag vehicle as unavailable	Vehicle excluded from bookings.
Notification	Trigger on key status changes	Customer/staff informed.
Reporting	Aggregate operational data	Management receives KPIs.

12.1 Rental Cost Calculation

```
long days = ChronoUnit.DAYS.between(pickupDate, returnDate);
if (days <= 0) {
    days = 1;
```

```
}
double total = dailyRate * days;
```

12.2 Payment Balance

```
double balance = totalAmount - amountPaid;

if (balance <= 0) {
    paymentStatus = "PAID";
} else {
    paymentStatus = "OUTSTANDING";
}
```

13. Automation Opportunities

Automation reduces repetitive work and helps the business respond consistently. The proposed system identifies several opportunities for automation without removing necessary human approvals.

Automation	Trigger	Action
Booking confirmation	Staff approves booking	Create confirmation and notification.
Availability update	Booking becomes confirmed	Reserve vehicle for selected dates.
Rental activation	Vehicle collected	Change status to RENTED.
Return processing	Vehicle returned	Record return and inspection.
Maintenance alert	Service date/mileage threshold reached	Notify fleet officer.
Payment reminder	Balance remains outstanding	Generate reminder.
Daily summary	Scheduled daily task	Generate dashboard summary.
Report generation	Manager requests report	Build report from database.
Audit logging	Important record changed	Store user, action and time.

13.1 Special Capabilities

- Vehicle availability calendar
- Search by registration, model, customer or booking number
- Role-specific dashboard
- Automatic rental price calculation
- Maintenance alerts
- Outstanding-balance monitoring
- Five-year historical archive
- Audit trail for important actions

14. Dashboard, Reporting and Social-Media Connectivity

14.1 Dashboard

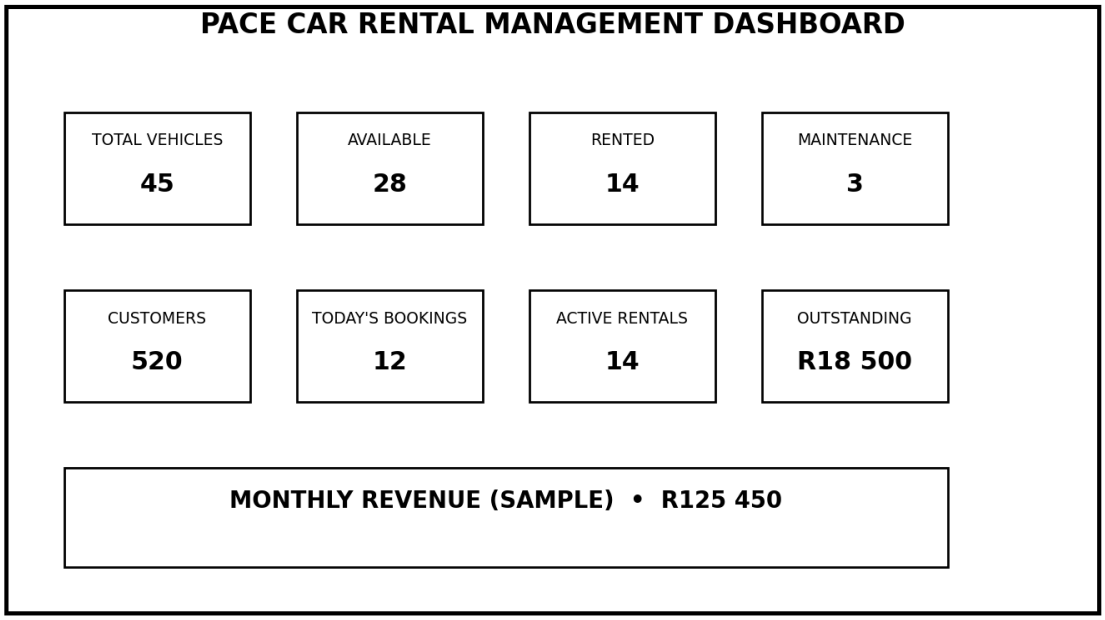


Figure 8: Sample management dashboard using fictional demonstration data

The dashboard is the central console for authorised management users. It should present current operational indicators without exposing unnecessary personal information. Managers can use it to understand vehicle utilisation, bookings, active rentals, payments and maintenance.

14.2 Proposed Reports

- Daily bookings report
- Monthly bookings report
- Vehicle availability report
- Vehicle utilisation report
- Revenue report
- Outstanding payment report
- Customer activity report
- Maintenance report
- Rental history report
- Cancelled booking report

14.3 Social-Media Connectivity

The system can provide a controlled marketing/contact area that links customers to the client's approved social-media channels. Where an official API is available and the client authorises access, selected functions such as publishing approved promotions or sharing booking-related content can be implemented. For the academic prototype, public links and mock integration points are sufficient; the system should not claim access to private accounts without client authorisation.

14.4 Browser Access

PICRMS is designed as a responsive web application. Customers and staff use a modern browser to access the appropriate pages. The application server handles authentication, business logic and database operations behind the browser interface.

15. Security Management

Security is a core requirement because the platform processes personal information, identity references, booking history and payment information. The system should follow the principle of least privilege: each user receives only the access required to perform their role.

15.1 Authentication

- Unique user accounts
- Strong password policy
- Password hashing rather than plain-text storage
- Session management
- Automatic session timeout
- Account lockout or rate limiting after repeated failures

15.2 Authorisation

Role	Access
Administrator	Users, roles, configuration, audit and all authorised management functions
Manager	Dashboard, reports, fleet and financial oversight
Rental Staff	Customers, bookings, rentals and approved payment functions
Fleet Officer	Vehicles, inspections and maintenance
Customer	Own profile, bookings and approved payment information

15.3 Data Protection

- Encrypt data in transit using HTTPS/TLS in production
- Restrict database credentials to the application service
- Use parameterised queries/JPA to reduce SQL-injection risk
- Validate all user inputs
- Avoid exposing sensitive data in browser responses
- Use audit logs for important changes
- Back up the database regularly
- Test restoration procedures
- Separate development, testing and production credentials

15.4 Audit Trail

The audit log should record important administrative and transactional actions, such as creating a user, changing a vehicle status, cancelling a booking or editing payment information. Audit records support accountability and troubleshooting.

16. Expected Benefits and Value to the Client

Business need	PICRMS value
Centralised information	One database provides a consistent view of operations.
Faster customer service	Staff can find customers, vehicles and bookings quickly.
Vehicle visibility	Fleet status and maintenance are visible in one module.
Booking accuracy	Date-overlap rules reduce conflicting reservations.
Financial control	Payments, balances and invoices are linked to bookings.
Management insight	Dashboard and reports provide operational KPIs.
Security	Roles and audit trails reduce unauthorised access.
Record retention	Historical records remain available for at least five years.
Automation	Notifications, calculations and status changes reduce repetitive work.
Scalability	Modular Jakarta EE design can support additional branches/services later.

16.1 Success Indicators

- Reduced time required to find booking information
- Fewer booking conflicts
- Improved accuracy of rental calculations
- Higher percentage of transactions with complete records
- Faster report preparation
- Improved visibility of vehicles under maintenance
- Successful recovery from tested backups
- Positive user feedback during pilot testing

17. Conclusion

The Pace Integrated Car Rental Management System is proposed as a unified platform for managing vehicle operations, customers, bookings, rentals, payments and maintenance. The three core aspects are deliberately connected through a central database and shared business rules, allowing information to flow through the rental lifecycle rather than being stored in isolated applications.

The proposed architecture uses Java and Jakarta EE to provide an enterprise application layer, with MySQL as the relational database and a responsive browser interface for customers and employees. The design includes authentication, role-based access, data validation, audit logging, backups, five-year record retention, automation, reporting and a central dashboard.

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